

Machine Learning

CIA – 3 Assignment

Domain: Health

Social Problem Statement:

Mental illness among tech-industry employees is common but frequently goes unaddressed, since stigma, fear of career repercussions, and uncertainty about workplace support discourage people from seeking treatment even when they need it. Without early identification of who is unlikely to seek help, employers and mental health advocacy organizations cannot target outreach effectively, and at-risk individuals remain unsupported until distress escalates.

Intended Beneficiaries:

Tech employees who are struggling but haven't sought treatment (primary beneficiaries, via earlier awareness and targeted resources); HR and wellness teams at tech companies, who can use aggregate risk patterns to design better mental health benefits and reduce stigma; and mental health advocacy nonprofits like OSMI, who can prioritize outreach and policy campaigns based on which workplace factors most suppress help-seeking.

Prediction target:

Binary classification: whether a survey respondent has sought mental health treatment (treatment: yes/no), predicted from demographic features (age, gender, country), workplace factors (company size, remote status, employer-provided mental health benefits, ease of taking medical leave), and attitudinal factors (family history, perceived stigma, comfort discussing mental health with supervisors/coworkers).

Measurable impact:

Model performance: ROC-AUC, F1, recall (prioritized, since failing to flag someone who needs support is costlier than a false positive), and precision, compared against a logistic regression/decision tree baseline. Downstream impact: identification of the workplace factors most strongly associated with non-treatment-seeking (via SHAP), which advocacy groups or HR teams could act on — e.g., quantifying how much perceived stigma or lack of benefits suppresses help-seeking, to prioritize interventions.

ML Justification:

Treatment-seeking behavior depends on many interacting, non-linear factors — stigma perception, benefits availability, family history, and workplace culture don't combine additively, and their relative importance varies across demographic subgroups. Tree-based ensembles (Random Forest) and stacking can capture these interactions better than a single rule-based heuristic, and the dataset is fully labeled, enabling supervised learning. The output is naturally probabilistic, suiting a risk-flagging or awareness-campaign-targeting use case rather than a hard yes/no diagnosis.

Dataset source and unit of analysis:

The OSMI (Open Sourcing Mental Illness) Mental Health in Tech Survey, collected via an anonymous self-report survey of tech-industry employees.

<https://www.kaggle.com/datasets/osmi/mental-health-in-tech-survey>

Unit of analysis: one individual survey respondent per row, capturing their demographics, employer's mental health policies, and personal/family mental health history.

Responsible-use limitations:

The data is self-reported and self-selected (people willing to complete a voluntary mental-health survey may not represent the broader tech workforce), skews toward US-based, English-speaking respondents, and is several years old, so results may not generalize across geography, company culture, or time. The model must never be used to identify or flag specific individuals for employer action — its role is limited to aggregate pattern discovery for advocacy/policy purposes. Fairness across gender, age, and country subgroups must be checked, since a model that performs unevenly could reinforce existing disparities in whose mental health needs get recognized. Finally, "treatment-seeking" is a proxy, not a diagnosis — it reflects access and willingness, not actual mental health status, and should never be framed as a clinical assessment tool.

Citations:

[1] "Mental Health Analysis of Employees using Machine Learning Techniques," in *Proc. IEEE Conf.*, 2021. doi: 10.1109/document/9668526.

[2] "Mental health in tech workplace: An analysis," *ResearchGate*, 2023. [Online]. Available: https://www.researchgate.net/publication/373976244_Mental_health_in_tech_workplace_An_analysis

[3] "Survey dataset on mental health in tech professionals from open sourcing mental health surveys (2017–2021)," *Data in Brief*, ScienceDirect, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352340924003469>

[4] "Quantum Machine Learning with HQC Architectures using non-Classically Simulable Feature Maps," *arXiv preprint*, arXiv:2103.11381. [Online]. Available: <https://arxiv.org/pdf/2103.11381>

Results and Figures

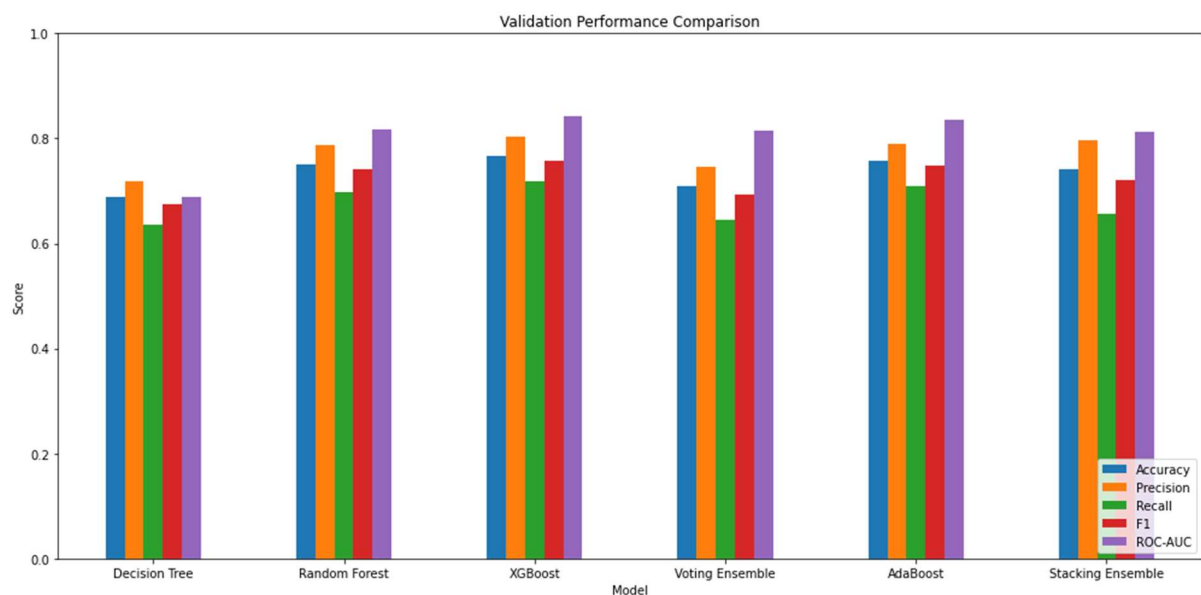
Results

The Decision Tree was used as the baseline, followed by Random Forest as the bagging model and XGBoost as the boosting model. A heterogeneous Voting Ensemble and Stacking Ensemble were also evaluated. XGBoost was selected as the best model based on validation F1. Its tuned configuration used 100 estimators, maximum depth 2, learning rate 0.03, subsample=1.0, and colsample_bytree=0.8, achieving a 5-fold cross-validation F1-score of approximately **0.766**.

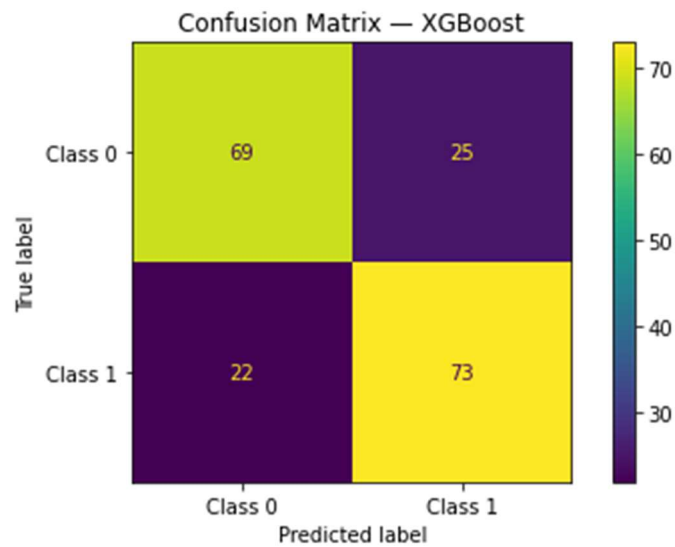
On the final test set of **189 observations**, the Decision Tree achieved 64% accuracy and 0.667 F1, while Random Forest achieved 74% accuracy and approximately 0.75 F1. XGBoost performed best with **75% accuracy, 0.745 precision, 0.768 recall, and 0.756 F1-score**. The Voting Ensemble and Stacking Ensemble performed below XGBoost, with accuracies of approximately 68% and 71%, respectively.

Most importantly, your explicit baseline comparison shows that XGBoost improved the Decision Tree's F1-score from **0.667 to 0.756**, an improvement of **0.090**, while also improving accuracy from 0.640 to 0.751 and ROC-AUC from 0.640 to 0.807.

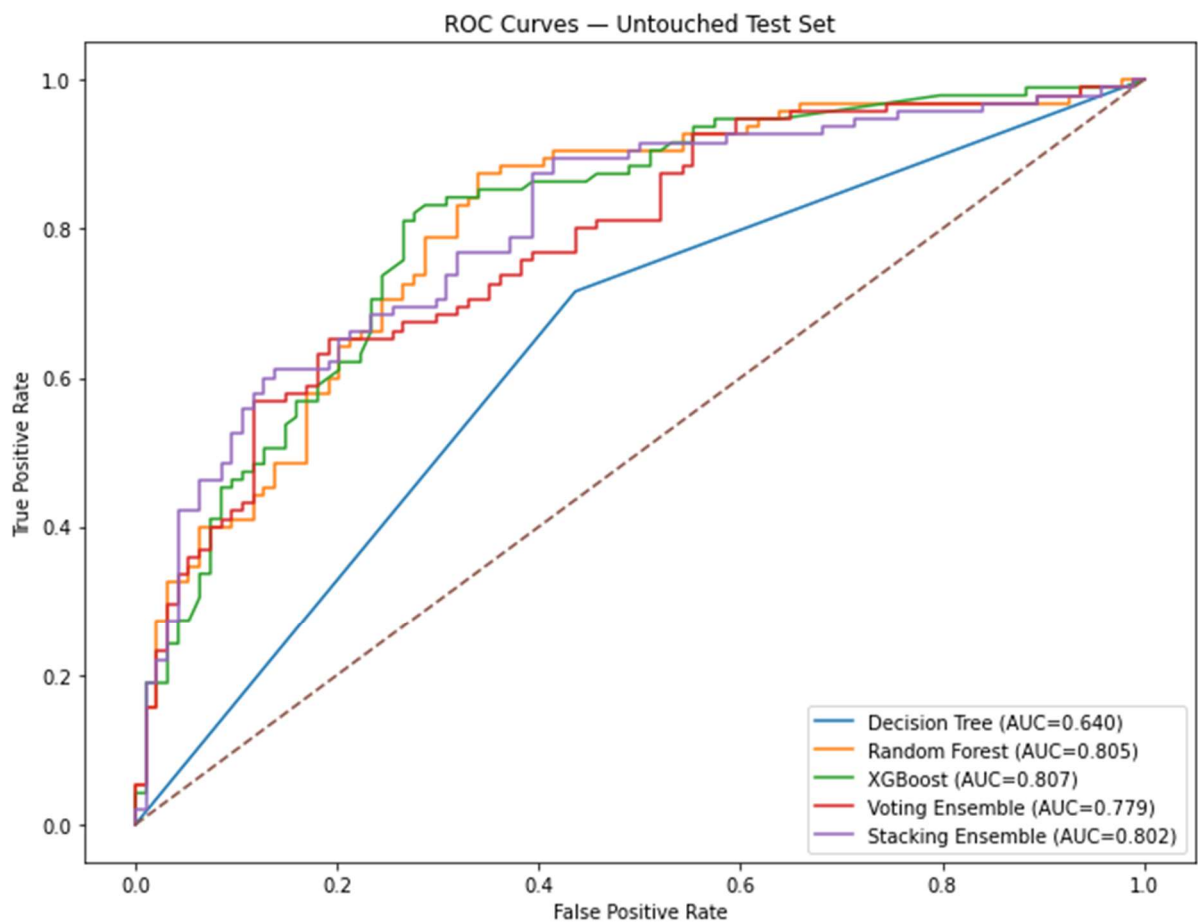
Figures



Comparison of classification performance across the baseline, bagging, boosting and ensemble models on the evaluation set.



Confusion matrix of the best-performing XGBoost model showing correct and incorrect classifications.

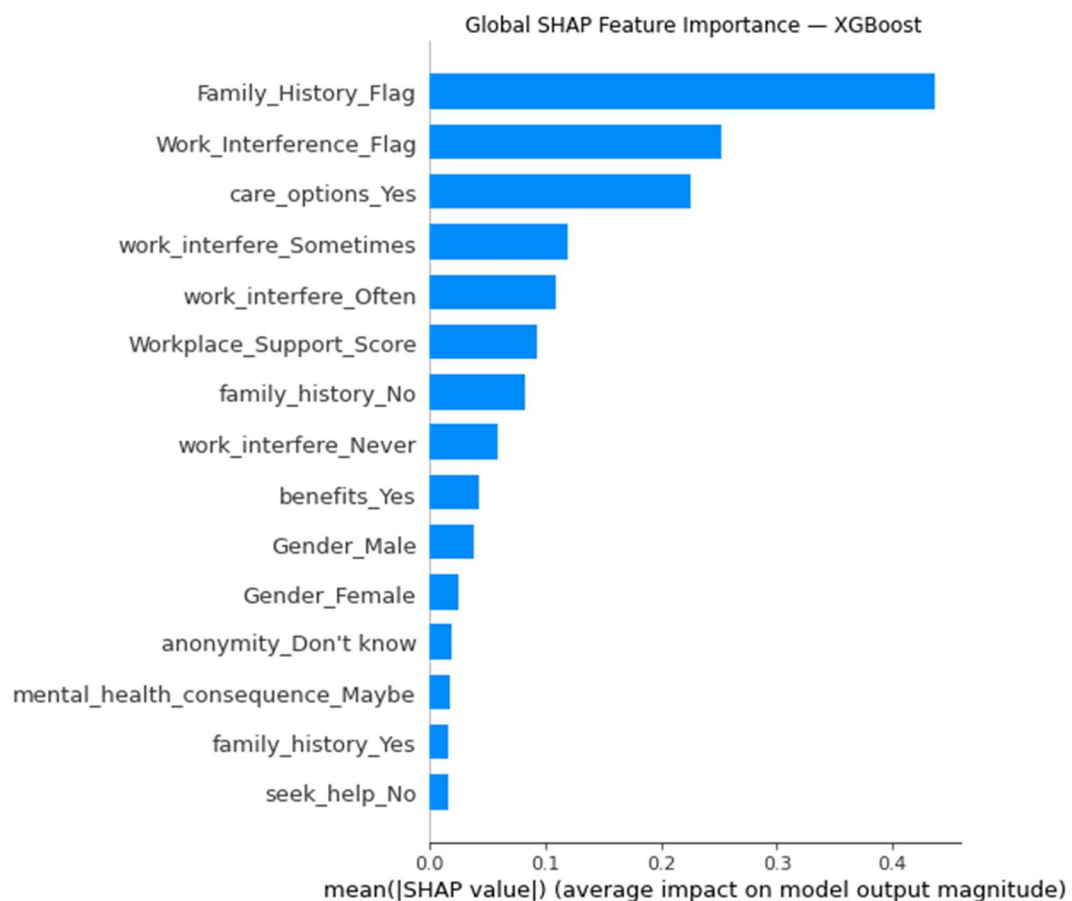


ROC curves comparing the classification performance of the evaluated models.

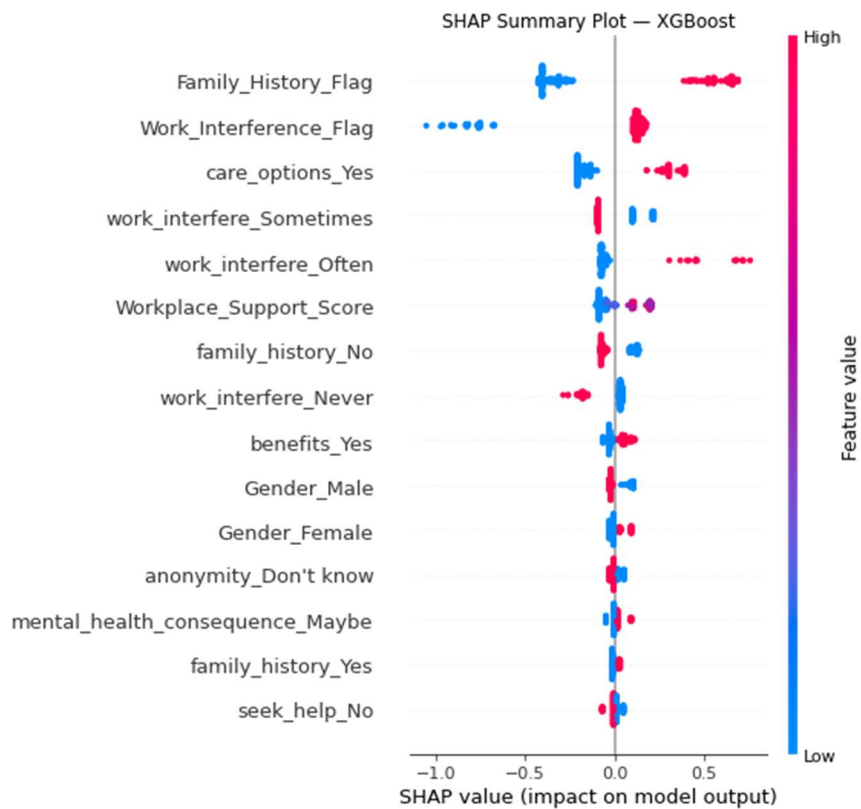
Explainability Output

SHAP was used to provide global and individual-level explanations for the final XGBoost model. The global SHAP feature-importance analysis ranks features according to their average absolute contribution to predictions, allowing the model's most influential variables to be identified. The SHAP summary plot additionally shows whether individual feature values push predictions toward or away from the positive class. Your notebook successfully generated the XGBoost global SHAP importance plot.

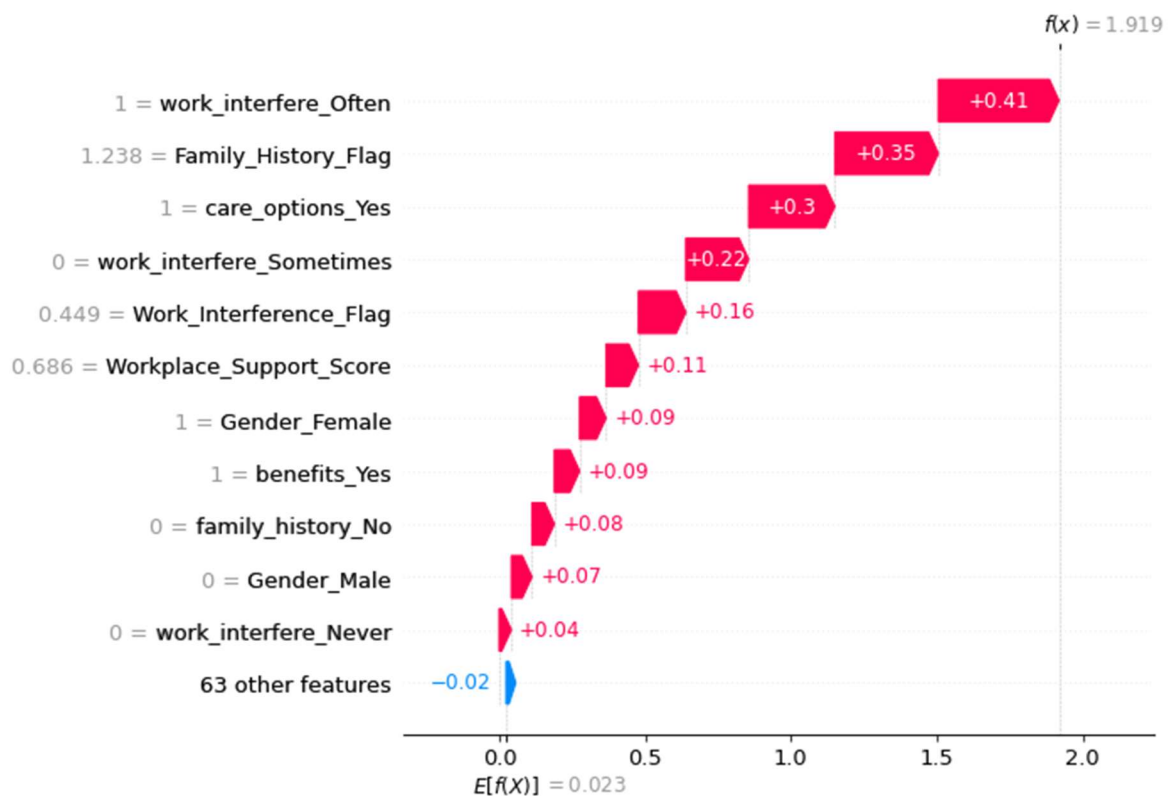
For individual-level explainability, the notebook creates a synthetic employee profile and passes it through the final XGBoost model. The model predicts "Treatment likely" with an 87.20% probability, compared with a 12.80% probability of no treatment prediction. The output explicitly notes that this is a model prediction rather than a diagnosis. The local SHAP analysis then ranks the features contributing to this particular prediction and identifies whether each factor increases or decreases the treatment prediction.



Global SHAP feature importance showing the features that most strongly influence XGBoost predictions.



SHAP summary plot showing the magnitude and direction of feature contributions to XGBoost predictions.



Local SHAP waterfall explanation for a synthetic individual prediction, showing the features that increased or decreased the predicted probability.